© 2026. Published by The Company of Biologists | Journal of Cell Science (2026) 139, jcs264960. doi:10.1242/jcs.264960

SHORT REPORT

SPECIAL ISSUE
IMAGING CELL ARCHITECTURE AND DYNAMICS

Tubulin polyglutamylation modulates Golgi morphodynamics and neurite branching during neuronal morphogenesis

Chih-Hsuan Hsu*, Alexander Josiah Kinrade* and Sarah Cohen‡

ABSTRACT

Neurons establish functional networks through morphological remodeling during neuronal differentiation. Microtubule polyglutamylation is a key microtubule post-translational modification that is highly enriched during this process and plays an important role in differentiation. However, how remodeling of organelle features such as morphology, distribution and interactions depends on tubulin polyglutamylation during neuronal differentiation remains unclear. Here, we employed multispectral imaging combined with quantitative three-dimensional organelle analysis to comprehensively profile eight organelles simultaneously in human induced pluripotent stem cell-derived neurons. We discovered that depletion of tubulin polyglutamylation induces pronounced alterations in somatic Golgi morphology and associated organelle interactions. In addition, Golgi-derived compartments in proximal neurites exhibited altered morphology and dynamics, namely, decreased anterograde transport together with less efficient movement. These changes were accompanied by increased neurite branching and tortuosity. Together, our findings reveal a previously unrecognized role for tubulin polyglutamylation in coordinating organelle organization with neurite architecture, providing a mechanistic link between tubulin post-translational modifications, Golgi morphology, dynamics and neuronal morphogenesis.

KEY WORDS: Cell biology, Microscopy, iNeurons, Microtubules, Polyglutamylation, Organelles, Golgi

INTRODUCTION

During neuronal differentiation, neurons undergo a series of morphological changes to establish highly polarized cellular architecture, including axons and dendrites, to form functional neural networks (Ho and Gupton, 2021; Menon and Gupton, 2018). These morphological changes require coordinated remodeling of the cytoskeleton, plasma membrane and intracellular organelles (Hanus and Ehlers, 2016; Koppers and Farías, 2021). As a major cytoskeletal component, microtubules play crucial roles in this process by providing structural support, serving as tracks for intracellular transport, and governing organelle organization and membrane trafficking (Burute and Kapitein, 2019; Schmoranzer and Simon, 2003). Microtubule dynamics and functions are regulated by diverse tubulin isotypes and post-translational modifications (PTMs), collectively known as the 'tubulin code' (Janke and Magiera, 2020; Roll-Mecak, 2020), which has been shown to play important roles in neuronal development (Hsu et al., 2026 preprint; Park and Roll-Mecak, 2018). Tubulin polyglutamylation is one of the key tubulin PTMs that markedly accumulates in neurons during differentiation (Audebert et al., 1994; Hsu et al., 2026 preprint). It can function as a signal to direct organelle positioning, particularly of the endoplasmic reticulum (ER), and thereby potentially influence the distribution of other organelles through membrane contact sites (Zheng et al., 2022).

Studies have shown that metabolic reprogramming accompanies neuronal differentiation, a process often associated with changes in organelle organization and function, especially for mitochondria (Ordureau et al., 2021; Zanellati et al., 2026 preprint; Zhang et al., 2022; Zheng et al., 2016). In parallel, extensive multi-organelle rearrangement occurs across selective autophagy and secretory trafficking, involving coordinated changes among the ER, Golgi, lysosomes and mitochondria (Hoyer et al., 2024; Ordureau et al., 2021; Wang et al., 2023). Our recent studies have shown that tubulin PTMs such as tubulin acetylation coordinate organelle organization and are important for maintaining organelle homeostasis during neuronal differentiation (Hsu et al., 2026 preprint). Although tubulin polyglutamylation and organelle remodeling have both been implicated in neuronal differentiation, no systematic analysis has been performed to understand how tubulin polyglutamylation contributes to the global landscape of multi-organelle remodeling to support the development of complex neuron architecture.

Multispectral imaging has increasingly been applied as a systems-level approach for multi-organelle analysis, enabling comprehensive characterization of organelle features at the single-cell level (Arribat et al., 2020; Hsu et al., 2026 preprint; Rhoads et al., 2025; Valm et al., 2017; Zanellati et al., 2026 preprint; Zheng et al., 2022; Zimmermann et al., 2024). This approach overcomes the spectral channel limitations of conventional fluorescence imaging while avoiding the low-throughput, labor-intensive preparation and harsh fixation required for electron microscopy, and retains the speed and sensitivity of confocal microscopy (Zanellati et al., 2024). Here, we applied multispectral imaging together with an in-house computational analysis pipeline called 'infer-subc' (https://github.com/SCohenLab/infer-subc/tree/v2.0.0b1) to examine how eight organelles remodel in response to depletion of tubulin polyglutamylation in human induced pluripotent stem cell (hiPSC)-derived neurons (iNeurons). We discovered that tubulin polyglutamylation predominantly affects the somatic Golgi among the organelles examined, accompanied by altered organelle interactions

Department of Cell Biology and Physiology, University of North Carolina at Chapel Hill, Chapel Hill, NC 27599, USA.
*These authors contributed equally to this work

‡Author for correspondence (sarahcoh@med.unc.edu)

C.-H.H., 0000-0002-5599-807X; A.J.K., 0009-0006-5813-3289; S.C., 0000-0003-2982-3265

This is an Open Access article distributed under the terms of the Creative Commons Attribution License (https://creativecommons.org/licenses/by/4.0), which permits unrestricted use, distribution and reproduction in any medium provided that the original work is properly attributed.

Handling Editor: Francesca Bottanelli
Received 13 April 2026; Accepted 17 September 2026

involving the Golgi. In addition, tubulin polyglutamylation depletion changes the dynamics of Golgi-derived compartments in neurites. We further found that these alterations are associated with increased neurite branching. Together, these findings reveal a previously uncharacterized mechanism linking tubulin polyglutamylation-dependent regulation of organelle organization to neurite morphology.

RESULTS AND DISCUSSION

Multispectral imaging reveals organelle remodeling in response to tubulin polyglutamylation during neuronal differentiation

Our recent work demonstrated that coordinated remodeling of multiple organelles is a hallmark of neuronal differentiation (Zanellati et al., 2026 preprint). To study organelle remodeling during neuronal differentiation, we used a hiPSC NGN2 (also known as NEUROG2) differentiation system to generate cortical neurons (iNeurons) in a controlled and rapid manner (Fig. 1A). We previously found that during neuronal differentiation, tubulin polyglutamylation levels become highly enriched and show preferential subcellular accumulation in proximal neurites compared with the soma (Hsu et al., 2026 preprint) (Fig. 1B). Tubulin polyglutamylation is catalyzed by tubulin tyrosine ligase-like (TTLL) enzymes that add glutamate residues to the C-terminal tails of tubulin. TTLL1 serves as a primary α-tubulin polyglutamylase responsible for extending glutamate side chains on microtubules and is highly expressed in the brain (Janke et al., 2005; van Dijk et al., 2007). To investigate how tubulin polyglutamylation regulates organelle networks during neuronal differentiation, we reduced polyglutamylated tubulin levels by lentiviral shRNA-mediated knockdown of TTLL1. shRNA was transduced immediately after NGN2 induction (day 0), and cells were collected at day 7, achieving efficient knockdown as verified by western blotting (Fig. 1C) and immunofluorescence staining (Fig. 1D).

Next, we implemented a systematic multispectral microscopy technique to comprehensively capture three-dimensional (3D) organelle organization under conditions of tubulin polyglutamylation perturbation. iNeurons treated with non-targeting (shCtrl) versus TTLL1-targeting (shTTLL1) shRNA were labeled with markers for eight organelles – endoplasmic reticulum (ER), peroxisomes (PO), Golgi (GL), mitochondria (MC), lysosomes (LS), plasma membrane (PM), lipid droplets (LD) and nucleus (NU) – and imaged at day 7 to visualize organelle organization (Fig. 1E,F). Single timepoint *z*-stacks were acquired, linearly unmixed and deconvolved, followed by instance segmentation of organelles in each channel (Fig. S1A,B). The expression level of the fluorescent reporters was unchanged between knockdown versus control groups (Fig. S1C). We analyzed organelle features related to morphology, inter-organelle communication and spatial distribution using the infer-subc pipeline, generating 5439 metrics, from which a curated subset of 383 was selected for downstream analysis. We found 28 metrics that were significantly different between shCtrl- and shTTLL1-treated day 7 iNeurons at a false discovery rate of 10%, of which more than 78% of identified metrics were associated with Golgi-related changes, based on the sialyltransferase (SiT) reporter (Fig. 1G; Table S1). As perturbation could potentially affect SiT localization, we performed immunofluorescence staining of the endogenous cis- and trans-Golgi markers GM130 (also known as GOLGA2) and Golgin-97 (also known as GOLGA1). Airyscan imaging showed that both compartments appeared dispersed, suggesting a disruption of overall Golgi organization (Fig. S2A). Therefore, for simplicity, we refer to these phenotypes as Golgi changes. Overall, we identified that depleting tubulin polyglutamylation resulted in distinct organelle remodeling, primarily in the soma and largely associated with Golgi-related features.

Tubulin polyglutamylation-dependent organelle remodeling predominantly involves somatic Golgi

The curated 3D organelle multispectral imaging analysis revealed that the most significant changes observed were in Golgi morphology in the soma of shCtrl- versus shTTLL1-treated day 7 iNeurons. We identified an increase in the number of Golgi structures, while the size of individual Golgi structures decreased, resulting in an overall unchanged total Golgi volume and reduced Golgi size variability (Fig. 2A,B). In parallel, the Golgi shape became more fragmented, less irregular and with fewer elongations, as indicated by decreased major axis length and equivalent diameter, and increased extent and solidity (Fig. 2C). Besides, GM130-labeled cis-Golgi changes also showed similar changes in soma (Fig. S2B). In addition to Golgi, lysosome morphology was also significantly altered, with changes such as decreased individual lysosome size, reduced major axis length and equivalent diameter, indicating a shift toward smaller lysosomes (Fig. 2B,C). Notably, distinct tubulin PTMs exhibit differential effects on organelle remodeling, especially in lysosome morphology upon depletion of tubulin-modifying enzymes. In contrast to depleting tubulin polyglutamylation, tubulin acetylation depletion results in enlarged lysosomes (Hsu et al., 2026 preprint). As these organelle organization changes were identified using the same multispectral imaging approach, this comparison highlights both the specificity of individual tubulin modifications in shaping organelle architecture and the sensitivity of the method to resolve distinct subcellular phenotypes. Next, we quantified lateral (*xy*) and axial (*z*) organelle distributions, representing the spread of organelles from the nucleus to the cell border and along the vertical axis of the cell from the bottom to the top, respectively. The lateral distribution of the Golgi remained unchanged, whereas the Golgi showed increased variability in axial distribution in shTTLL1-treated iNeurons (Fig. 2D). To reduce the possibility of shTTLL1 (KD1) off-target effects, we tested another independent shTTLL1 (KD2), which yielded a similar trend toward Golgi fragmentation, based on GM130 morphology (Fig. S2C). Moreover, expressing an shTTLL1-resistant TTLL1 increased polyglutamylated tubulin levels, reflecting the restoration of polyglutamylase activity, and partially restored Golgi size, providing further support of the specificity of the observed phenotype (Fig. S2D).

Our recent work demonstrates that remodeling of inter-organelle communication networks rewires lipid metabolism to support neuronal differentiation (Zanellati et al., 2026 preprint). However, no studies have systematically characterized how tubulin polyglutamylation mediates these inter-organelle interactions during neuronal differentiation. To address this, we quantified the overlapping regions between segmented organelle objects, referred to as organelle interactions as previously described (Hsu et al., 2026 preprint; Rhoads et al., 2025; Zanellati et al., 2026 preprint). Among all examined Golgi–organelle pairs, Golgi–lysosome interactions were the most abundant. However, Golgi–peroxisome interactions showed the largest changes upon depleting tubulin polyglutamylation, with increases in both the number and volume of interactions (Fig. 2E, left). The overall spatial distribution of interactions was not significantly altered, except for Golgi–lysosome interactions, which shifted in the axial direction (Fig. 2E, middle). Nevertheless, Golgi–peroxisome interactions became more randomly distributed in both

Journal of Cell Science (2026) 139, jcs264960. doi:10.1242/jcs.264960

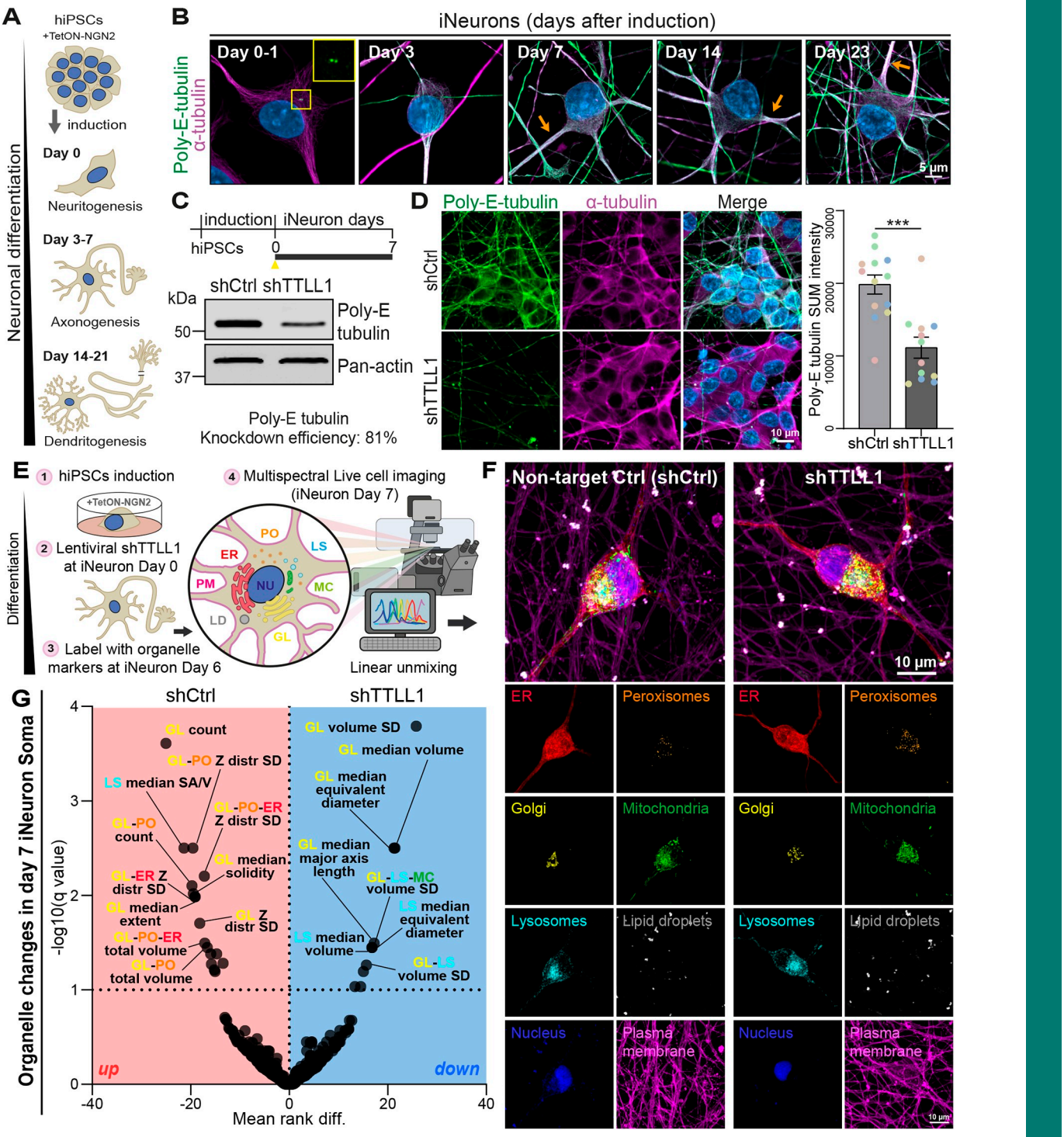


Fig. 1. See next page for legend.

the lateral and axial directions (Fig. 2E, right). One possible explanation is that Golgi fragmentation and dispersal increase opportunities for Golgi–peroxisome interactions. In addition, Golgi–peroxisome interplay has been implicated in supporting Golgi lipid composition and secretory function (König et al., 2025 preprint). Thus, altered Golgi architecture might increase reliance on Golgi–peroxisome interactions as a compensatory response to maintain Golgi lipid composition and support secretory demands. Analysis of three-way organelle interactions revealed the largest changes in interactions involving Golgi and peroxisomes. Whereas Golgi–peroxisome–lysosome interactions were the most abundant, Golgi–peroxisome–ER interactions showed significant increases in both interaction number and volume with minimal spatial changes in shTTLL1-treated iNeurons (Fig. 2F).

In summary, depletion of polyglutamylated tubulin resulted in prominent Golgi remodeling, characterized by increased Golgi fragmentation and greater variability in axial distribution. In addition to these morphological and spatial changes, Golgi–

Journal of Cell Science

Fig. 1. Multispectral imaging reveals organelle remodeling in response to tubulin polyglutamylation during neuronal differentiation. (A) Schematic, adapted from Hsu et al. (2026 preprint), illustrating the NGN2-induced neuronal differentiation process across distinct morphogenesis stages. TetON-hNGN2-KOLF2.1J human induced pluripotent stem cells (hiPSCs) were differentiated into cortical neurons (iNeurons) driven by NGN2 induction. (B) Airyscan *z*-stack confocal images with maximum-intensity projections of methanol-fixed iNeurons at days 0–1, 3, 7, 14 and 23 post induction. Cells were immunolabeled for polyglutamylated tubulin (green) and α-tubulin (magenta). Nuclei were labeled with DAPI (blue). Orange arrows mark the proximal neurites. *n*=1 independent experiments. (C) Top: schematic diagram of non-target shRNA (shCtrl) versus TTLL1 shRNA (shTTLL1) lentivirus transduction and cell lysate collection timelines, where the yellow arrowhead marks the day of viral transduction and the bar denotes the duration from transduction to sample collection. Bottom: western blot analysis shows correlating polyglutamylated tubulin protein levels. *n*=1 independent experiments. (D) Representative immunofluorescence confocal images show shCtrl- versus shTTLL1-treated day 7 iNeurons fixed and stained for polyglutamylated tubulin (green), α-tubulin (magenta) and DNA (DAPI, blue). Bar graph shows quantification of tubulin polyglutamylation sum-projected signal intensity across the whole field of view, normalized to cell number per field of view, from five independent cultures (color coded), demonstrating decreased tubulin polyglutamylation in shTTLL1- compared with shCtrl-treated iNeurons (shCtrl, total of 151 cells; shTTLL1, total of 129 cells). Error bars represent±s.e.m. Unpaired two-tailed *t*-test. **P*<0.05. (E) Schematic of multispectral imaging workflow, adapted from Hsu et al. (2026 preprint). ER, endoplasmic reticulum; GL, Golgi; LD, lipid droplets; LS, lysosomes; MC, mitochondria; PM, plasma membrane; PO, peroxisomes. (F) Individual channels from processed 3D multispectral images display eight organelles separately. (G) Volcano plot illustrating soma-specific differences in 3D organelle organization (383 metrics) between shCtrl- and shTTLL1-treated day 7 iNeurons. Statistical significance was determined using a 10% false discovery rate threshold. diff., difference; distr, distribution; SA/V, surface area to volume ratio; SD, standard deviation; Z, projected imaging depth along *z*-axis. *n*=39 shCtrl- and 46 shTTLL1-treated cells from three independent cultures.

peroxisome interactions increased and became more randomly distributed. Moreover, interactions among Golgi, peroxisomes and ER were also enhanced. As both peroxisomes and ER play central roles in lipid metabolism, increased Golgi interactions with these organelles might reflect enhanced coordination to meet lipid demands required for membrane trafficking.

Reducing tubulin polyglutamylation alters trans-Golgi morphology and dynamics in neurites

The somatic Golgi plays a central role in neuronal secretory trafficking and primary dendrite specification, whereas Golgi-derived compartments within proximal dendrites act as local trafficking hubs that contribute to microtubule nucleation, neurite branching and migration (Nakagawa, 2024; Ori-McKenney et al., 2012). Polyglutamylated microtubules are highly enriched in proximal neurites compared with the soma (Fig. 1B), raising the possibility that local tubulin polyglutamylation depletion affects the organization of these Golgi-derived compartments in proximal neurites. No significant changes were observed in the 122 neurite-specific curated subset among organelles examined (Fig. S3A; Table S1). However, this might be due to the strong somatic SiT signals that limit the detection of dim SiT-positive structures in neurites in the multispectral imaging scheme. Examination of the cis- and trans-Golgi markers GM130 and Golgin-97 showed an increase in GM130 structures within shTTLL1 neurites positive for the dendritic marker MAP2, whereas the overall Golgin-97 amount was unchanged (Fig. S3B), suggesting that somatic Golgi fragmentation is not correlated with trans-Golgi abundance in proximal neurites. We therefore next asked whether tubulin polyglutamylation depletion instead alters the dynamics of Golgi-derived compartments in proximal neurites.

SiT was expressed alone, followed by live-cell tracking within proximal neurites (Fig. 3A, Movie 1). Depletion of tubulin polyglutamylation reduced SiT anterograde movement (Fig. 3B). The decreased speed and confinement ratio while increasing total turning angle indicate slower and less directionally persistent transport. Although the median value of track duration remained unchanged, the mean track duration increased, suggesting an increased proportion of longer-lasting tracks in shTTLL1-treated neurites (Fig. 3C). To enrich faster-moving particles consistent with microtubule-based transport, we further classified tracks as sustained fast tracks (mean track speed ≥0.7 µm/s) or transient fast tracks (maximum track speed ≥0.7 µm/s). Both fast-track groups showed a shift toward retrograde-biased movement following polyglutamylated tubulin depletion (Fig. S3C).

In addition to dynamic changes, all three track populations showed consistent morphological alterations, including particles becoming smaller, rounder and more morphologically uniform (Fig. 3D; Fig. S3D), indicating a transition toward a more vesicular and transport-ready state in shTTLL1-treated neurites. Together, our findings suggest that somatic Golgi fragmentation following polyglutamylated microtubule depletion does not alter the overall trans-Golgi abundance in neurites. Instead, these smaller and rounder particles might be preferentially redirected or retained within proximal neurites, with reduced anterograde transport, pointing to an impact on local secretory pathways, including altered distribution of newly synthesized proteins and lipids.

Loss of tubulin polyglutamylation alters neuronal morphology and increases neurite branching

Small Golgi outposts were previously found to selectively partition into longer, more complex dendrites to support directed secretory trafficking required for asymmetric dendrite growth and morphogenesis (Horton et al., 2005). In addition, α-tubulin polyglutamylation, regulated by doublecortin (DCX), has been implicated in restricting neurite branching during early neuronal development, as DCX knockout iNeurons exhibit increased neurite initiation without major changes in microtubule dynamics (Sébastien et al., 2025). The altered SiT transport observed in neurites upon TTLL1 depletion suggested changes in local membrane delivery through secretory trafficking. Therefore, we next examined neuronal morphology and neurite network organization. At this stage, iNeurons expressed the dendritic marker MAP2, the pan-axonal marker labeling neurofilaments, and the axon initial segment marker ankyrin-G (also known as ANK3); iNeuron cultures were highly intertwined, making it challenging to distinguish neurites from individual cells (Fig. S4A). Cell masks generated from the multispectral dataset were leveraged, as sparse labeling enabled segmentation and quantification of individual neurons (Fig. 4A, left). Tubulin polyglutamylation depletion did not alter soma volume but significantly increased the number of primary neurites directly extending from the soma (Fig. 4A, right).

Next, to examine the effect of tubulin polyglutamylation on neurite network organization, iNeuron cultures were labeled with Cell Tracker Green and analyzed by skeletonization. Depleting polyglutamylated tubulin increased neurite tortuosity with decreased neurite width. The total neurite content per field of view was comparable, but tubulin polyglutamylation depletion resulted in a greater number of shorter and more fragmented

Journal of Cell Science (2026) 139, jcs264960. doi:10.1242/jcs.264960

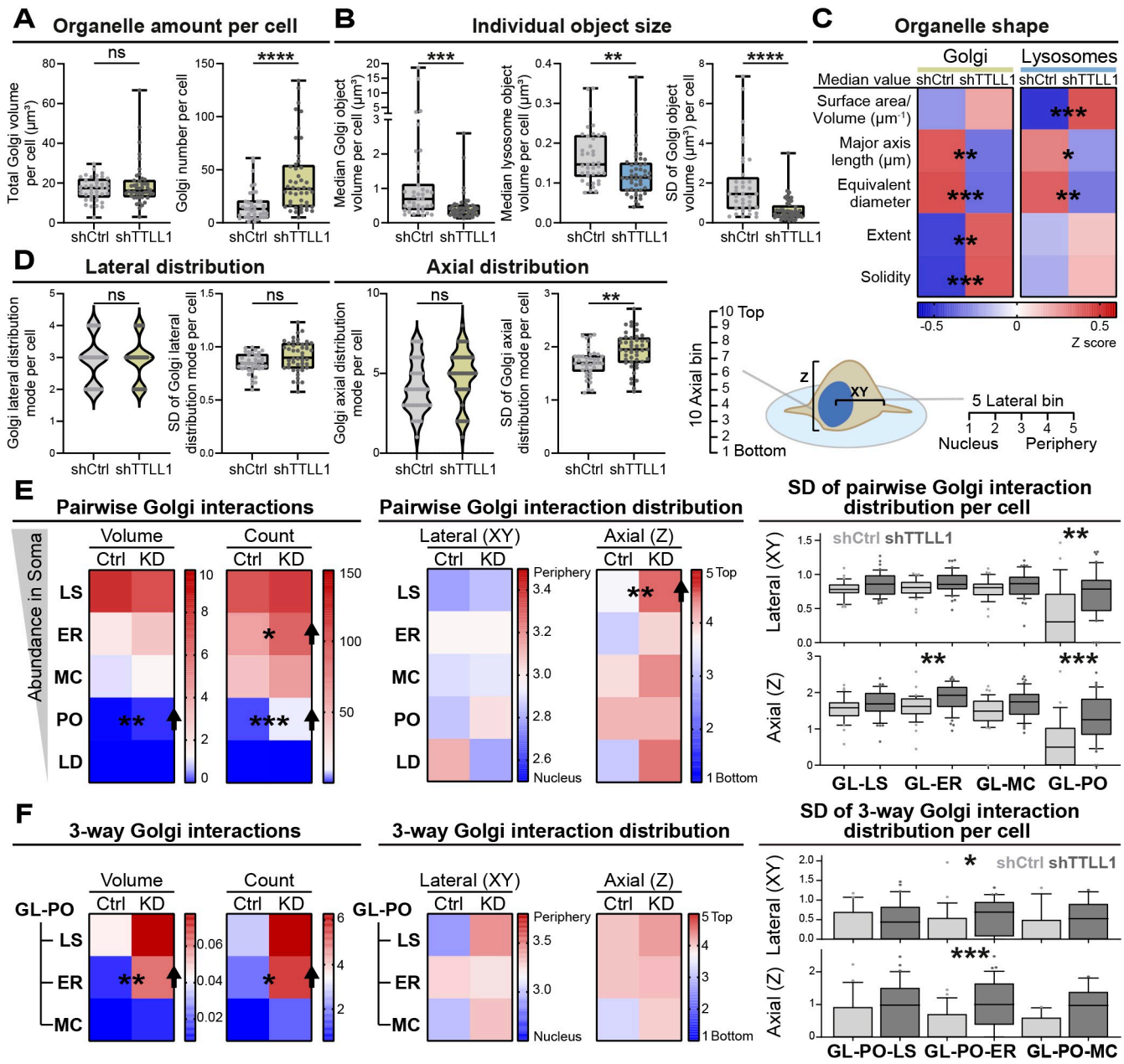


Fig. 2. Tubulin polyglutamylation-dependent organelle remodeling occurs predominantly involved Golgi. (A,B) Box plots comparing shCtrl- versus shTTLL1-treated day 7 iNeurons for Golgi total volume (A, left), Golgi object number (A, right), individual Golgi object median size (B, left), individual lysosome median size (B, middle), and standard deviation (SD) values of Golgi object volumes (B, right). Data points represent single cells. Asterisks denote *q*-values from Fig. 1G. (C) Heatmaps displaying the organelle shape metrics for Golgi and lysosomes. Each metric is Z-score-normalized with asterisks denoting *q*-values from Fig. 1G. (D) Schematic of organelle distribution measurements along the *xy*- and *z*-axes (right), adapted from Hsu et al. (2026 preprint). Quantification of the mode and SD values of Golgi across each dimension (left). Asterisks denote *q*-values from Fig. 1G. (E) Heatmaps of the total volume (µm^3) and number of pairwise soma-specific Golgi interactions per cell in shCtrl (Ctrl)- versus shTTLL1 (KD)-treated day 7 iNeurons, ordered by organelle volume abundance (left). Pairwise Golgi interaction distribution mode along the lateral and axial are displayed in heatmaps (middle), with box plot analysis of lateral or axial distribution SD values for each Golgi pairwise interaction (right). Asterisks indicate *q*-values from Fig. 1G. (F) Three-way interactions involving Golgi and peroxisomes are quantified by total volume (µm^3) and number (left), interaction distribution mode (middle) and SD of lateral and axial distributions (right) comparing shCtrl- and shTTLL1-treated day 7 iNeurons. Asterisks indicate *q*-values from Fig. 1G. (A–F) *n*=39 shCtrl- and 46 shTTLL1-treated cells from three independent cultures. For A,B,D, the box represents the 25–75th percentiles, the central line marks the median, and whiskers show minimum and maximum values. (E,F) Arrows indicate a significant increase (upward arrows) or decrease (downward arrows; did not occur) in the shTTLL1-treated day 7 iNeurons relative to the control condition. For E,F (right), the box shows the 25–75th percentiles, the central line marks the median, and whiskers show the 10–90th percentiles. ns, not significant; **q*<0.1; ***q*<0.05; ****q*<0.01; *****q*<0.001.

branches (Fig. 4B). To further characterize neurite networks, skeleton features, including endpoints, slabs and junctions, were analyzed. Although the number of slabs remained unchanged, polyglutamylated tubulin depletion resulted in an increased number of endpoints and junctions, indicating enhanced branching complexity (Fig. 4C).

In conclusion, multispectral imaging revealed that tubulin polyglutamylation mediated by TTLL1 regulates the morphology,

Journal of Cell Science (2026) 139, jcs264960. doi:10.1242/jcs.264960

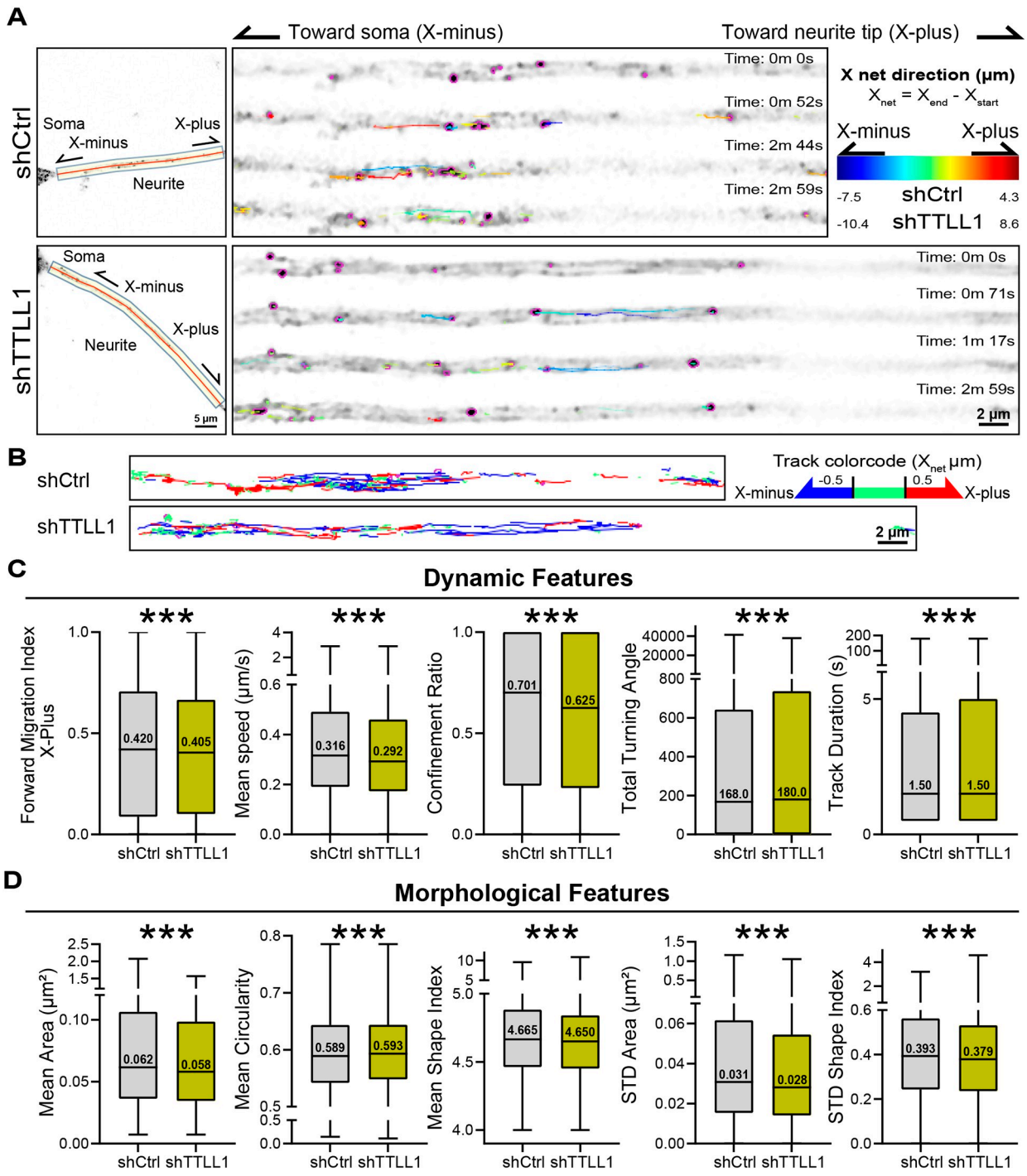


Fig. 3. Reducing tubulin polyglutamylation alters trans-Golgi morphology and dynamics in neurites. (A) Day 7 shCtrl- versus shTTLL1-treated iNeurons with sialyltransferase (SiT; trans-Golgi) labeling were imaged using Fast-Airyscan for 3 min at 0.5 s intervals. Neurite regions selected for analysis were photobleached corrected, tracked and straightened from the soma toward the distal neurite within the field of view, as indicated in the representative images. Tracks are color-coded by net movement in the *x*-direction (X_{net}). (B) Tracks from representative examples were color coded by net movement in the *x*-direction as >0.5 µm (red), between 0.5 µm and −0.5 µm (green), and < −0.5 µm (blue). (C,D) Box plots of sialyltransferase dynamics (C) and morphological features (D). STD, standard deviation. For track duration, mean±s.e.m. values are: shCtrl, 4.75±0.082 s; shTTLL1, 5.54±0.11 s. Boxes show the 25–75th percentiles, the central line marks the median, and whiskers show minimum and maximum values. Values represent population-level summaries across multiple neurites: *n* (neurites)=58 shCtrl (47 cells) and 55 shTTLL1 (39 cells) from four independent cultures. Asterisks denote *P*-values determined by a Bonferroni-corrected statistical randomization test. ***P<0.001.

distribution and interactions of the somatic Golgi among six major organelles, as well as the morphology and dynamics of Golgi-derived compartments in proximal neurites at an early differentiation stage (day 7). As prolonged tubulin polyglutamylation depletion compromised cell health, we could not determine whether the Golgi phenotype persists into more mature stages (Fig. S4B). Together with its elevated enrichment in proximal neurites, these findings support a spatially specific role

Journal of Cell Science (2026) 139, jcs264960. doi:10.1242/jcs.264960

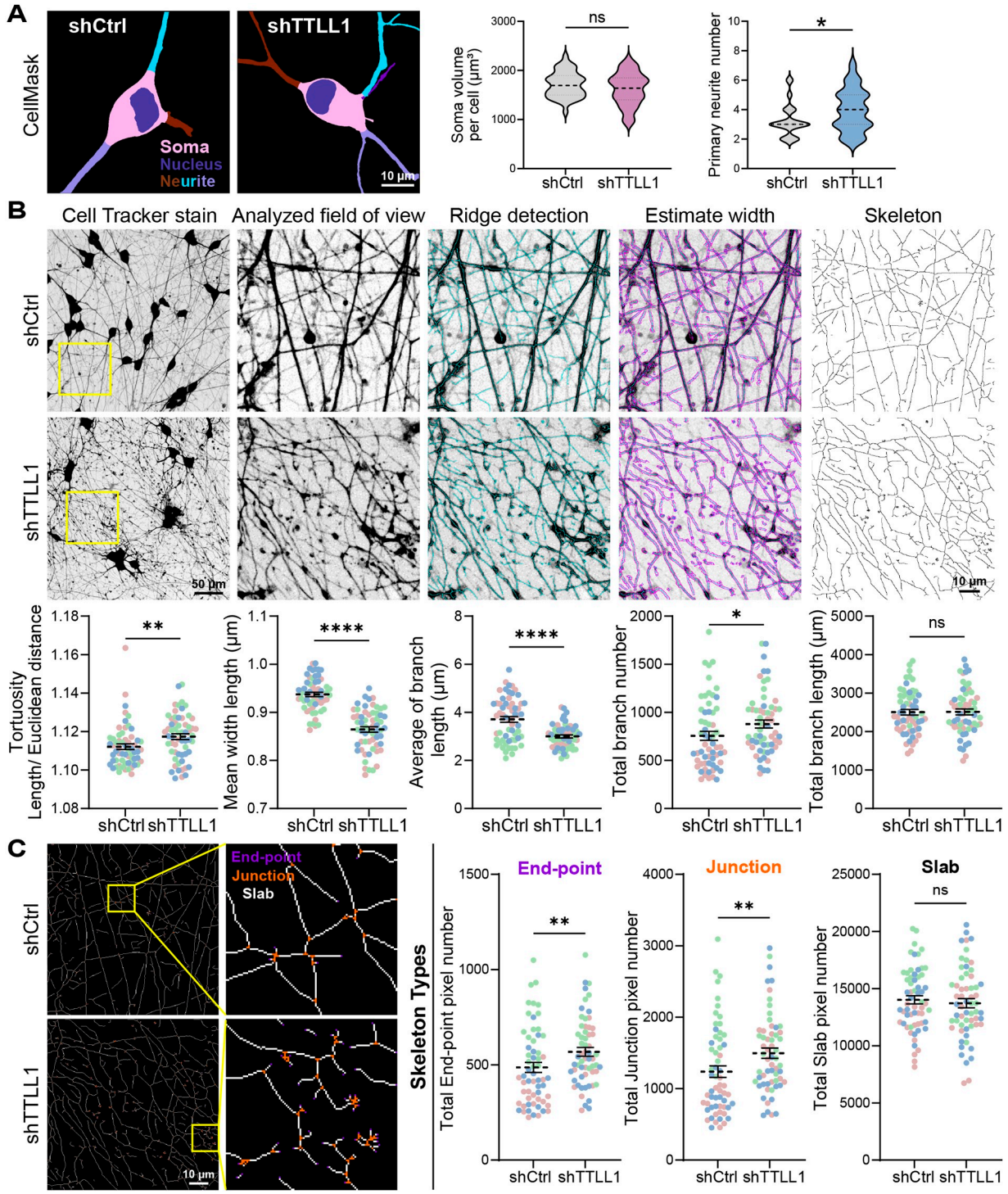


Fig. 4. Loss of tubulin polyglutamylation alters neuronal morphology and increases neurite branching. (A) Masks derived from multispectral processing were used to quantify soma volumes and primary neurite number between shCtrl- versus shTTLL1-treated day 7 iNeurons. *n*=39 shCtrl- and 46 shTTLL1-treated cells from three independent cultures. Statistical significance was determined using an unpaired two-tailed *t*-test. (B) Cell Tracker Green-labeled day 7 shCtrl- and shTTLL1-treated iNeurons were processed by ridge detection and skeletonization to quantify tortuosity index (ratio of skeletonized length to Euclidean distance), mean width length, average branch length, total branch number and total branch length. Data represent mean ±s.e.m. *n* (analyzed field of view)=60 shCtrl- and 60 shTTLL1-treated cells from three independent cultures, with data points color coded by independent cultures. Statistical significance for mean width length and total branch length was determined using an unpaired two-tailed *t*-test. For tortuosity index and average branch length, Mann–Whitney-test was applied. In the representative images, the yellow box indicates the cropped region for analysis, cyan lines mark detected neurite ridges, and magenta outlines estimated neurite width. (C) Skeletonized images from B were analyzed for pixel counts of end points, junctions and slabs (right). Mann–Whitney test was used for statistical analysis. ns, not significant; *P<0.05; **P<0.01; ****P<0.0001.

for polyglutamylated microtubules in controlling Golgi-derived compartment dynamics in neurites. The neuronal network phenotype we observed, characterized by increased branching and reduced straightness, might be explained by intracellular dysregulation linking tubulin polyglutamylation-dependent control of Golgi organization to neurite architecture (Fig. S4C).

MATERIALS AND METHODS

Culture of iNeurons

KOLF2.1J wildtype hiPSCs were obtained from Bill Skarnes (The Jackson Laboratory) and were used to generate a pBac-TetON-hNGN2 hiPSC stable cell line as previously described (Fernandopulle et al., 2018; Zanellati et al., 2026 preprint). Briefly, KOLF2.1J wildtype hiPSCs were transfected with 500 ng piggyBac plasmid encoding a bipartite TetON-hNGN2 cassette in a well of a six-well plate (plasmids gifted from Michael Ward, National Institute of Neurological Disorders and Stroke). Following transfection, cells were selected for stable genomic integration using puromycin (0.7 µg/ml). Cells were maintained and cultured as adherent monolayers at 37°C in a humidified incubator with 5% CO_2 under feeder-free conditions in StemFlex medium (A3349401, Gibco) on vitronectin (VTN-N)-coated six-well plates (VTN-N recombinant human protein, truncated; A14700, Gibco). Plates were coated with vitronectin by diluting 10 µl vitronectin in 1 ml of 1× Dulbecco's phosphate-buffered saline (DPBS; 14190144, Gibco) per well and incubating for 2 h at 37°C. The hiPSC medium was changed every 1–2 days and cells were passaged upon reaching ∼80% confluency using ReLeSR passaging reagent (100-0484, STEMCELL Technologies) according to the manufacturer's instructions in the presence of 0.5× ROCK inhibitor (RevitaCell Supplement; A2644501, Gibco).

Differentiation into iNeurons

An established protocol was adapted to differentiate pBac-TetON-hNGN2 hiPSCs into iNeurons (Fernandopulle et al., 2018; Pantazis et al., 2022) with modifications (Hsu et al., 2026 preprint; Zanellati et al., 2026 preprint). Cultured hNGN2 hiPSCs were dissociated into single cells using Accutase (A6964-100ML, Sigma-Aldrich) and cells [$(1.0–1.2)\times10^6$ per well of six-well plates] were seeded in induction medium (IM) consisting of KnockOut™ DMEM/F-12 (12660012, Gibco) supplemented with 1× N-2 (17502048, Gibco), 1× MEM non-essential amino acids solution (MEM-NEAA; 11140050× Gibco), 1× GlutaMAX (35050061, Gibco), 1× ROCK inhibitor and 0.5 µg/ml doxycycline hyclate (D9891, Sigma-Aldrich). 24 h after plating, the medium was replaced with IM (1× N-2, 1× MEM-NEAA, 1× GlutaMAX) with 1 µg/ml doxycycline hyclate. After 48 h, 4×10^4 induced cells were replated using Accutase onto each well of an eight-well Nunc Lab-Tek II chamber with 0.1 mg/ml poly-L-ornithine hydrobromide (P3655, Sigma-Aldrich) in borate buffer [10 mM boric acid (B6768, Sigma-Aldrich), 2.5 mM sodium tetraborate (221732, Sigma-Aldrich), 7.5 mM sodium chloride (S7653, Sigma-Aldrich), 0.1 M sodium hydroxide (71463, Sigma-Aldrich)] and 10 µg/ml laminin (Laminin Mouse Protein, Natural, 1 mg; 23-017-015, Gibco) coated in IM (1× N-2, 1× MEM-NEAA, 1× GlutaMAX) with 0.5 µg/ml doxycycline hyclate and 0.1× ROCK inhibitor. 72 h after doxycycline induction (defined as day 0 iNeurons), cultures were switched to cortical neuron culture medium consisting of BrainPhys (05791, STEMCELL Technologies), 10 µg/ml BDNF (450-02, PeproTech), 10 µg/ml NT-3 (450-03, PeproTech), 2% B-27 (A1486701, Gibco) and 1 µg/ml laminin, supplemented with 0.5 µg/ml doxycycline hyclate. iNeurons were cultured up to 23 days as adherent monolayers at 37°C in a humidified incubator with 5% CO_2 with half medium changes every 2–3 days in doxycycline-free cortical neuron culture medium.

iNeuron transduction

Lentiviral shRNA expression plasmids were from Sigma-Aldrich: MISSION® TRC-puro Non-Mammalian shRNA for control, TRCN0000048527 (KD1) and TRCN0000048524 (KD2) for TTLL1. All TTLL1 knockdown experiments were performed using KD1, except those in Fig. S2C, which were using KD2. Lentivirus was concentrated with Lenti-X Concentrator (631231, Takara Bio) in Milli-Q water and cells were treated with virus diluted 1:2000 in cortical medium. 24 h after the transduction, the medium was replaced with fresh cortical neuron culture medium.

Western blotting

Cells were lysed on ice in RIPA buffer (50 mM Tris, pH 8, 150 mM NaCl, 0.5% deoxycholate, 0.1% SDS, 1% NP-40 supplemented with 15 mM sodium pyrophosphate, 50 mM sodium fluoride, 40 mM β-glycophosphate, 1 mM sodium vanadate, 100 mM phenylmethylsulfonylfluoride, 1 µg/ml leupeptin and 5 µg/ml aprotinin) supplemented with 1:100 protease inhibitor cocktail (P8340-5mL, Sigma-Aldrich), 1:100 phosphatase inhibitor cocktail 2 (P5726, Sigma-Aldrich) and 1:100 phosphatase inhibitor cocktail 3 (P0044, Sigma-Aldrich). Adherent samples were scraped, collected and centrifuged at 13,000 ***g*** for 5 min at 4°C to isolate the post-nuclear supernatant. The concentration of the supernatant was measured by Pierce BCA Protein Assay Kits (23225, Thermo Fisher Scientific); the supernatants were then boiled with 6× Laemmli buffer containing β-mercaptoethanol and denatured for 10 min at 95°C. Equal protein amounts were loaded into each well of a 4–20% Mini-PROTEAN TGX Precast Protein Gel (Bio-Rad) and transferred onto 0.2 µm nitrocellulose membranes (162-0112, Bio-Rad) by wet transfer 1 h at 100 V for immunoblotting. Membranes were blocked in 5% milk in Tris-buffered saline (TBS) for 30 min at room temperature and then incubated with primary antibodies diluted in 5% milk in TBS containing 0.1% Tween 20 (TBST) overnight at 4°C. The next day, membranes were washed three times for 5 min each with TBST, incubated with secondary antibodies diluted in 5% milk in TBST for 1 h at room temperature, washed three times for 5 min each with TBST, and imaged using an Odyssey CLx imager (LI-COR Biosciences) (Fig. S5). Western blots were quantified using LI-COR Image Studio. Background was subtracted for each quantified band. Knockdown efficiency was calculated by normalizing the signal intensity of polyglutamylated tubulin to that of pan-actin and then comparing the normalized values between knockdown and control groups.

Antibodies

The antibodies used were: rabbit polyclonal anti-polyglutamate chain (polyE) [pAb (IN105); AdipoGen, AG-25B-0030, RRID:AB_2490540; 1:300 for immunofluorescence (IF), 1:5000 for western blotting (WB)], mouse anti-α-tubulin DM1A (Abcam, ab7291, RRID:AB_2241126; 1:1000 IF, 1:10,000 WB), mouse monoclonal against actin (SPM161; Santa Cruz Biotechnology, sc-56459, RRID:AB_830981; 1:5000 WB), chicken anti-MAP2 polyclonal (Thermo Fisher Scientific, PA1-10005, RRID:AB_1076848; 1:1000 IF), mouse anti-GM130 (BD Biosciences, 610822, RRID:AB_398141; 1:400 IF), rabbit polyclonal anti-Golgin-97 (Abcam ab84340, RRID:AB_1860512; 1:400 IF), mouse anti-neurofilament marker (pan axonal, cocktail; BioLegend, 837904, RRID:AB_2566782; 1:500 IF), mouse monoclonal anti-ankyrin-G (N106/36; Antibodies Incorporated, 75-146, RRID:AB_10673030; 1:150 IF), rabbit polyclonal anti-βIII tubulin (Abcam, ab18207, RRID:AB_444319; 1:500 IF) and chicken polyclonal anti-GFP (Aves Labs, GFP-1010, RRID:AB_2307313; 1:500 IF).

Immunofluorescence staining and imaging

Cells were fixed with either ice-cold 100% methanol for 10 min at −20°C or 4% paraformaldehyde (PFA) for 15 min at 37°C, then washed with DPBS three times. Cells were blocked using a blocking solution of 4% bovine serum albumin (BSA) in DPBS for 45–60 min. PFA-fixed cells were permeabilized with 0.2% Triton X-100 in DPBS for 10 min at room temperature and washed with DPBS. After fixation, primary antibodies were diluted in blocking solution and incubated with cells overnight at 4°C, and then cells were washed with DPBS in three intervals of 5 min. Secondary antibodies were diluted in blocking solution at room temperature for 1 h while covered from light. After incubation, the secondary antibodies were washed out using three 5-min DPBS washes. For additional nuclear staining, cells were incubated with DAPI (1:1000 in DBPS) for 10 min following secondary antibody incubation, then washed with DPBS once. Immunolabeled cells were imaged using a Zeiss laser scanning microscope (LSM) 800 with a 63×1.4 NA oil objective. For Fig. 1B and Fig. S2A,B, Airyscan *z*-slice images were collected and processed using ZEN 2.3

Airyscan 3D processing software. For staining shown in Fig. 1B,D, methanol-fixed samples were incubated with antibodies against polyglutamylated tubulin and α-tubulin DM1A. For staining shown in Fig. S2C, methanol-fixed samples were incubated with antibodies against polyglutamylated tubulin and GM130. For cis- and trans-Golgi staining shown in Figs S2A and S3B, methanol-fixed samples were incubated with antibodies against MAP2, GM130 and Golgin-97. For axon initial segment staining, methanol-fixed samples were incubated with an anti-ankyrin-G antibody, together with antibodies against MAP2 and βIII tubulin. For axon staining, PFA-fixed samples were incubated with a pan-axonal maker (neurofilament SMI312) antibody together with antibodies against MAP2 and βIII tubulin. For neuronal marker staining in Fig. S4A, cells were imaged using a Plan-Apochromat 20×/0.8 M27 objective.

Multispectral imaging of organelles

Day 6 iNeurons were co-transfected with six piggyBac organelle reporter plasmids together with a piggyBac transposase plasmid using Lipofectamine 2000 (11668027, Thermo Fisher Scientific) following a modified version of the manufacturer's protocol. For each well of an eight-well chamber, plasmid DNA (120 ng EIF1α-transposase, 120 ng LAMP1::mTurquoise for lysosomes, 120 ng Cox8::eGFP for mitochondria, 120 ng mApple::Sec61B for ER, 60 ng SiT::oxVenus for Golgi, 25 ng mOrange::SKL for peroxisomes, and 60 ng mTagBFP2::NLS for nuclei) was diluted in 25 µl neurobasal medium (12348017, Gibco). In parallel, 1 µl Lipofectamine 2000 was diluted in 25 µl neurobasal medium and incubated for 5 min at room temperature. The diluted DNA and Lipofectamine 2000 solutions were then combined, properly mixed and incubated for 15 min at room temperature. Prior to transfection, cultures were washed and equilibrated with neurobasal medium. After a 45-min incubation with the transfection mixture at 37°C and 5% CO_2, cultures were washed once and the medium was replaced with BrainPhys Imaging Optimized Medium (05796, STEMCELL Technologies) supplemented with 2% NeuroCult SM1 (05711, STEMCELL Technologies), 10 µg/ml BDNF, 10 µg/ml NT-3 and 1 µg/ml laminin. On day 7 of iNeuron differentiation, 15 min prior to live-cell imaging, 1:2000 CellMask Deep Red (C10046, Invitrogen) was added to label the plasma membrane and 0.5 µM Lipi-Blue (LD01; DOJINDO Laboratories) was added to label lipid droplets. For single-label controls, each plasmid or dye was applied individually using the same DNA-to-Lipofectamine ratios or dye concentrations as in multi-labeled conditions.

Multispectral imaging

Multispectral imaging was carried out as previously described (Hsu et al., 2026 preprint; Rhoads et al., 2025; Zanellati et al., 2026 preprint). Images were acquired using a Zeiss LSM 880 confocal microscope on an inverted Axio Observer Z1 base equipped with a 34-channel GaAsP spectral detector and a live-cell incubation chamber (Carl Zeiss Microscopy). 405 nm (0.5%), 458 nm (3%), 514 nm (0.4%), 561 nm (0.3%) and 633 nm (0.2%) lasers were used simultaneously to excite cells labeled with one or multiple organelle markers. Emitted fluorescence was collected in lambda mode using a linear array of 34 photomultiplier tubes with 9 nm spectral bins spanning 410–695 nm, resulting in 32 intensity channels per image. *Z*-stack images were acquired by a plan-Apo 63×/1.4 NA oil-immersion objective with an *xyz* voxel size of 0.07 µm×0.07 µm×0.39 µm. *Z*-stacks were acquired at a single time point with a pixel density of 796×796 per slice, and each *z*-slice was captured in 1.97 s. The number of *z*-slices per image was determined on a cell-by-cell basis to capture the entire cell volume from basal to apical surfaces. All imaging was performed live and cells were maintained at 37°C and 5% CO_2 during imaging.

Multispectral image processing

Using the linear unmixing algorithm from ZEN Black software version 2.3 (Carl Zeiss Microscopy), spectral *z*-stack images were first processed into eight-channel images. Control images labeled with a single fluorophore corresponding to one of the eight organelle markers were acquired as described above and used to generate reference spectra. For each fluorophore, a representative region of fluorescence intensity was selected to define its reference spectrum. All reference spectra were then used as inputs and applied to all images across the full biological replicate dataset for linear unmixing, resulting in an eight-channel image in which each channel represents fluorescence intensity from a single fluorophore.

Linear unmixed images were subsequently deconvolved using Huygens Essential software version 23.04 (Scientific Volume Imaging, The Netherlands, http://svi.nl). Excitation and emission maxima for each fluorophore were estimated from the lasers used during imaging and the corresponding reference spectra, and applied for batch deconvolution of eight intensity channels using the Workflow Processor module.

For lateral (*xy*) distribution, segmented *z*-stacks were summed across *z*-planes and divided into five equal bins with region 1 as the nucleus and regions 2–5 spanning from the nuclear edge to the cell periphery. For axial (*z*) distribution, summed projections across *xy*-axes were divided into ten equal bins from bottom to top (1 to 10) to quantify organelle volumes within each *z*-section.

3D multispectral image segmentation

Instance organelle segmentations were generated from independent intensity channels in the deconvolved *Z* stack images. Detailed segmentation methods are described in the image analysis pipeline in our Python-based segmentation and analysis package, infer-subc (https://github.com/SCohenLab/infer-subc/tree/v2.0.0b1). Lysosome and mitochondria object masks were the only organelles to be processed using the optional declumping step that exists at the end of each organelle segmentation workflow. Whole-cell masks (CellMask) and nuclear masks were generated using segmentation workflow 1.1c, followed by manual refinement in Napari based on plasma membrane and ER signals. CellMasks were subsequently processed using a dedicated workflow to segment the soma and neurites into separate soma and neurite masks (segmentation workflow 1.8).

3D multispectral organelle quantification

Quantitative analysis of 3D organelle organization was performed in infer-subc (https://github.com/SCohenLab/infer-subc/tree/v2.0.0b1). Following segmentation, individual organelle objects and inter-organelle interactions were quantified. Organelle interactions were defined as regions of overlap between organelle objects and were assessed for all combinations among the six organelles. For each organelle and organelle interaction, features including morphology (amount, volume and shape) and spatial distribution were extracted. Analyses were performed from the full cellular volume captured in each image which was further divided into soma and neurite subregions. All metrics were quantified per-subregion using the infer_subc analysis pipeline ('organelle-signature-analysis.ipynb').

Re-expression of TTLL1 in shTTLL1 iNeurons

The shRNA-resistant-TTLL1 (piggyBac-rTTLL1-IRES-eGFP) construct was generated based on the shTTLL1 TRCN0000048527 (KD1) target sequence by introducing silent mutations within the shTTLL1-binding site. Day 1 iNeurons were transfected with piggyBac-rTTLL1-IRES-eGFP plasmid along with a piggyBac transposase plasmid using Lipofectamine LTX (A12621, Invitrogen) following a modified version of the manufacturer's protocol. For each well of an eight-well chamber, plasmid DNA (100 ng EIF1α-transposase and 300 ng piggyBac-rTTLL1-IRES-eGFP) combined with 0.1 µl/100 ng of PLUS Reagent (A12621, Invitrogen) was diluted in 25 µl neurobasal medium. In parallel, 0.5 µl Lipofectamine LTX was diluted in 25 µl neurobasal medium. The diluted DNA and Lipofectamine LTX solutions were incubated for 5 min before combining, properly mixing and incubating for an additional 15 min. Prior to transfection, cells were washed and equilibrated in neurobasal medium. The combined mixture was then added dropwise to the cells and incubated for 45 min at 37°C and 5% CO_2 before being washed with neurobasal medium and replaced with cortical neuron culture medium. PFA-fixed and permeabilized cells were immunostained with antibodies against GFP, GM130 and polyglutamylated tubulin. To assess polyglutamylation levels following re-expression of shRNA-resistant TTLL1 in Fig. S2D, *z*-stack images were sum-intensity projected, processed using Subtract Background (rolling ball radius=50), and the mean gray values of individual somatic regions were measured.

Immunofluorescence-based Golgi morphology imaging and analysis

For TTLL1-depleted and re-expressing assays in Fig. S2C,D, *z*-stacks of cells were collected using a Zeiss LSM 800 microscope with a 64×1.4 NA oil objective. *Z*-stack images were sum-intensity projected, processed using Subtract Background (rolling ball radius =50), and regions of interest were cropped to the field of view in Fiji, then inputted in CellProfiler for batch quantification. Briefly, the cis-Golgi channel (GM130) was intensity rescaled, with noise reduction (size=5, distance=5 and cutoff distance=0.05), then a Gaussian filter (sigma=1), followed by segmentation. After generating the object masks, the object number and mean object area were quantified. All conditions were quantified using an identical CellProfiler pipeline with fixed parameters, applied in batch mode.

For cis- and trans-Golgi compartment analysis in Figs S2A and S3B, the soma regions were identified using an intensity-rescaled MAP2 channel and manually segmented to identify the regions of interest. Briefly, both the GM130 and Golgin-97 channels were intensity-rescaled and segmented (GM130: adaptive, minimum cross-entropy, lower bound=0.15, upper bound=1.0, window size=30; Golgin-97: adaptive, Otsu, lower bound=0.14, upper bound=1.0, window size=15). After generating object masks, the object number and object area were quantified. All conditions were quantified using an identical CellProfiler pipeline with fixed parameters, applied in batch mode.

The neurite regions were identified using an intensity-rescaled MAP2 channel and manually segmented to identify the regions of interest. Briefly, both the GM130 and Golgin-97 channels were intensity-rescaled and segmented (GM130: adaptive, minimum cross-entropy, lower bound=0.45, upper bound=1.0, window size=30; Golgin-97: adaptive, Otsu, lower bound=0.09, upper bound=1.0, window size=15). After generating object masks, the object number and object area were quantified. All conditions were quantified using an identical CellProfiler pipeline with fixed parameters, applied in batch mode.

Live-cell assay of trans-Golgi dynamics in neurites

Day 6 iNeurons were transfected with a single piggyBac organelle reporter plasmid with a piggyBac transposase plasmid using Lipofectamine 2000 following a modified version of the manufacturer's protocol. For each well of an eight-well chamber, plasmid DNA (30 ng EIF1α-Transposase and 120 ng SiT::oxVenus) was diluted in 25 µl neurobasal medium. In parallel, 1 µl of Lipofectamine 2000 was diluted in 25 µl neurobasal medium. The diluted DNA and Lipofectamine 2000 solutions were then combined, properly mixed and incubated for 15 min at room temperature. Prior to transfection, cells were washed and equilibrated with neurobasal medium. After a 45 min incubation with the transfection mixture at 37°C and 5% CO_2, cultures were washed once with neurobasal medium and replaced with cortical neuron culture medium. Airyscan timelapse images were acquired on a Zeiss LSM 880 microscope with a plan-Apo 63×/1.4 NA oil-immersion objective and processed using ZEN 2.3 Airyscan 2D Processing software. Timelapse images included a single *z*-plane imaged every 0.5 s for 3 min (360 frames). Cells were imaged live and maintained at 37°C and 5% CO_2 during imaging. The dynamics of trans-Golgi vesicles were quantified from 2D timelapse datasets using TrackMate and CellTracksColab (Ershov et al., 2022; Gómez-de-Mariscal et al., 2024). Timelapse images were processed prior to analysis. Briefly, neurite segments were isolated using the straighten function in Fiji and trans-Golgi vesicle signals were enhanced by correcting photobleaching (simple ratio, background=20), subtracting background signal (radius=8), employing a median filter (radius=1) and using the enhance contrast function (saturate pixels=0.35%) prior to vesicle segmentation. In TrackMate, the threshold detector (intensity threshold=40,000) and LAP tracker (linking max distance=1.5, area penalty=1.0, Y penalty=10.0; gap closing max distance=5.0, max frame gap=2, radius penalty=3.0, Y penalty=15.0) were used. The CSV files from TrackMate were imported into CellTracksColab to visualize and analyze the tracking data. CellTracksColab v1.0.4 was used following the recommended data structure and analysis steps for the raw tracks. Metrics of trans-Golgi vesicle dynamics calculated by TrackMate include circularity, confinement ratio and shape index. Circularity is the measure of how closely an object resembles a circle, whereas shape index is the measure of object irregularity and compactness. Confinement ratio describes the directness of a track by comparing the net distance traveled to the total displacement. The metrics from CellTracksColab analysis included: Forward Migration Index X-Minus, Forward Migration Index X-Plus and Total Turning Angle. Forward Migration Index X-Minus and X-Plus quantify the fraction of movement within a track directed in the negative and positive *x*-directions, respectively. Total Turning Angle is the sum of the angles between each pair of consecutive direction vectors along the track.

Neurite morphology assay

To visualize neuronal morphology, day 7 iNeurons were incubated with Cell Tracker Green (Invitrogen, C7025) (25 µM) for 15 min at 37°C and washed once with complete medium prior to imaging. Following dye incubation, *z*-stacks of cells were collected using a Zeiss LSM 800 microscope with a 20× objective. *Z*-stack images were maximum-intensity projected and then regions of interest were randomly selected, excluding soma-containing areas, and cropped to the field of view for analysis. Images were then processed using the Enhance Local Contrast (CLAHE) plugin in Fiji, followed by median filtering (radius=2). Processed images were analyzed using the Ridge Detection plugin in Fiji to detect neurite ridges and estimate width (Steger, 1998; https://doi.org/10.5281/zenodo.594213). Ridge-detected images were then converted to skeletonized images and analyzed using the Analyze Skeleton (2D/3D) plugin in Fiji (Arganda-Carreras et al., 2010; Doube et al., 2010; Polder et al., 2010).

Brightfield imaging for cell morphology

For images in Fig. S4B, brightfield images were acquired using an EVOS M5000 inverted microscope (Thermo Fisher Scientific) equipped with a transmitted-light illumination module using a 20× objective.

Statistics

The number of replicates and statistical tests are indicated in the figure legends. In figure legends, *n* represents the number of individual cells analyzed and pooled from independent experiments. For comparisons between two groups, two-tailed unpaired *t*-tests or Mann–Whitney-test were used. For multiple-group comparisons, one-way ANOVA followed by appropriate post hoc tests (specified in the figure legends) was performed. For microscopy experiments, fields of view were selected at random, but image acquisition and analysis were not performed masked to experimental conditions. For multispectral imaging datasets, a nonparametric Mann–Whitney *U*-test and statistical significance was determined by a 10% false discovery rate (Figs 1G and 2A–F). All statistical analyses, plots and heatmaps were conducted using GraphPad Prism 11 (RRID:SCR_002798; GraphPad Software, Boston, MA, USA; www.graphpad.com). For trans-Golgi dynamic tracking, the statical significance was determined using the Bonferroni-corrected statistical outcomes from the CellTracksColab randomization test (Gómez-de-Mariscal et al., 2024).

Acknowledgements

We thank Dr Bill Skarnes (The Jackson Laboratory) for generously providing the KOLF2.1J cell line. We thank Dr Michael Ward (National Institute of Neurological Disorders and Stroke) for kindly sharing piggyBac vectors. We thank Wendy Salmon [Hooker Imaging Core, University of North Carolina (UNC) School of Medicine] for expert imaging advice. We thank Shannon Rhoads, Ngudiankama Rene Mfulama and Zachary Coman for assistance with the 3D organelle analysis pipeline (infer-subc). We thank Maria Clara Zanellati for helping with the neuron differentiation protocol. Multispectral microscopy (Zeiss LSM 880 confocal microscope) was performed at the UNC Hooker Imaging Core Facility, supported in part by the P30 CA016086 Cancer Center Core support grant to the UNC Lineberger Comprehensive Cancer Center.

Competing interests

The authors declare no competing or financial interests.

Author contributions

Conceptualization: C.-H.H., S.C.; Formal analysis: C.-H.H., A.J.K.; Investigation: C.-H.H., A.J.K.; Resources: C.-H.H., S.C.; Supervision: S.C.; Writing – original draft: C.-H.H.; Writing – review and editing: C.-H.H.; A.J.K.; S.C.

Funding
This work was supported by the National Institutes of Health/National Institute of General Medical Sciences under the award R35GM133460 (to S.C.), the Chan Zuckerberg Initiative award A23-0264-001 (to S.C.) and by the Ministry of Education, Taiwan, under the Government Scholarship to Study Abroad (GSSA) program (to C.-H.H.). Open Access funding provided by University of North Carolina at Chapel Hill. Deposited in PMC for immediate release.

Data and resource availability
The data generated in this study are available from the corresponding author upon reasonable request. Multispectral imaging data are available at BioImage Archive under accession number S-BIAD3209. All other relevant data and details of resources can be found within the article and its supplementary information.

First Person
This article has an associated First Person interview with Chih-Hsuan Hsu, one of the co-first authors of the paper.

Special Issue
This article is part of the Special Issue 'Imaging Cell Architecture and Dynamics', guest edited by Francesca Bottanelli and Giulia Zanetti. See related articles at https://journals.biologists.com/jcs/issue/139/21.

References

Arganda-Carreras, I., Fernández-González, R., Muñoz-Barrutia, A. and Ortiz-De-Solorzano, C. (2010). 3D reconstruction of histological sections: application to mammary gland tissue. *Microsc. Res. Tech.* **73**, 1019-1029. doi:10.1002/jemt.20829

Arribat, Y., Grepper, D., Lagarrigue, S., Qi, T., Cohen, S. and Amati, F. (2020). Spastin mutations impair coordination between lipid droplet dispersion and reticulum. *PLoS Genet.* **16**, e1008665. doi:10.1371/journal.pgen.1008665

Audebert, S., Koulakoff, A., Berwald-Netter, Y., Gros, F., Denoulet, P. and Eddé, B. (1994). Developmental regulation of polyglutamylated α- and β-tubulin in mouse brain neurons. *J. Cell Sci.* **107**, 2313-2322. doi:10.1242/jcs.107.8.2313

Burute, M. and Kapitein, L. C. (2019). Cellular logistics: unraveling the interplay between microtubule organization and intracellular transport. *Annu. Rev. Cell Dev. Biol.* **35**, 29-54. doi:10.1146/annurev-cellbio-100818-125149

Doube, M., Kłosowski, M. M., Arganda-Carreras, I., Cordelières, F. P., Dougherty, R. P., Jackson, J. S., Schmid, B., Hutchinson, J. R. and Shefelbine, S. J. (2010). BoneJ: free and extensible bone image analysis in ImageJ. *Bone* **47**, 1076-1079. doi:10.1016/j.bone.2010.08.023

Ershov, D., Phan, M.-S., Pylvänäinen, J., Rigaud, S., Le Blanc, L., Charles-Orszag, A., Conway, J., Laine, R., Roy, N., Bonazzi, D. et al. (2022). TrackMate 7: integrating state-of-the-art segmentation algorithms into tracking pipelines. *Nat. Methods* **19**, 829-832. doi:10.1038/s41592-022-01507-1

Fernandopulle, M. S., Prestil, R., Grunseich, C., Wang, C., Gan, L. and Ward, M. E. (2018). Transcription factor–mediated differentiation of human iPSCs into neurons. *Curr. Protoc. Cell Biol.* **79**, e51. doi:10.1002/cpcb.51

Gómez-de-Mariscal, E., Grobe, H., Pylvänäinen, J. W., Xénard, L., Henriques, R., Tinevez, J.-Y. and Jacquemet, G. (2024). CellTracksColab is a platform that enables compilation, analysis, and exploration of cell tracking data. *PLoS Biol.* **22**, e3002740. doi:10.1371/journal.pbio.3002740

Hanus, C. and Ehlers, M. D. (2016). Specialization of biosynthetic membrane trafficking for neuronal form and function. *Curr. Opin. Neurobiol.* **39**, 8-16. doi:10.1016/j.conb.2016.03.004

Ho, C. and Gupton, S. L. (2021). Neurons | establishing and maintaining neuron morphology. *Encycl. Biol. Chem. III Third Ed.* **5**, 345-357. doi:10.1016/B978-0-12-819460-7.00240-1

Horton, A. C., Rácz, B., Monson, E. E., Lin, A. L., Weinberg, R. J. and Ehlers, M. D. (2005). Polarized secretory trafficking directs cargo for asymmetric dendrite growth and morphogenesis. *Neuron* **48**, 757-771. doi:10.1016/j.neuron.2005.11.005

Hoyer, M. J., Capitanio, C., Smith, I. R., Paoli, J. C., Bieber, A., Jiang, Y., Paulo, J. A., Gonzalez-Lozano, M. A., Baumeister, W., Wilfling, F. et al. (2024). Combinatorial selective ER-phagy remodels the ER during neurogenesis. *Nat. Cell Biol.* **26**, 378-392. doi:10.1038/s41556-024-01356-4

Hsu, C.-H., Kinrade, A. J., Zanellati, M. C. and Cohen, S. (2026). Tubulin acetylation governs organelle remodeling and lysosomal reformation during neuronal differentiation. *bioRxiv*, 2026.02.13.705749. doi:10.64898/2026.02.13.705749

Janke, C. and Magiera, M. M. (2020). The tubulin code and its role in controlling microtubule properties and functions. *Nat. Rev. Mol. Cell Biol.* **21**, 307-326. doi:10.1038/s41580-020-0214-3

Janke, C., Rogowski, K., Wloga, D., Regnard, C., Kajava, A. V., Strub, J.-M., Temurak, N., van Dijk, J., Boucher, D., van Dorsselaer, A. et al. (2005). Tubulin polyglutamylase enzymes are members of the TTL domain protein family. *Science* **308**, 1758-1762. doi:10.1126/science.1113010

König, M. A., Kucharowski, N., Ramos, D. P. D., Soyka, H., Wunderling, K., Bülow, T. R., Yaghmour, M. H., Thiele, C., Ache, J. M., Kuerschner, L. et al. (2025). Insulin-like peptide secretion is mediated by peroxisome-Golgi interplay. *bioRxiv*, 2025.05.26.656179. doi:10.1101/2025.05.26.656179

Koppers, M. and Farías, G. G. (2021). Organelle distribution in neurons: logistics behind polarized transport. *Curr. Opin. Cell Biol.* **71**, 46-54. doi:10.1016/j.ceb.2021.02.004

Menon, S. and Gupton, S. (2018). Recent advances in branching mechanisms underlying neuronal morphogenesis. *F1000Res.* **7**, F1000 Faculty Rev-1779. doi:10.12688/f1000research.16038.1

Nakagawa, N. (2024). The neuronal Golgi in neural circuit formation and reorganization. *Front. Neural Circuits* **18**, 1504422. doi:10.3389/fncir.2024.1504422

Ordureau, A., Kraus, F., Zhang, J., An, H., Park, S., Ahfeldt, T., Paulo, J. A. and Harper, J. W. (2021). Temporal proteomics during neurogenesis reveals large-scale proteome and organelle remodeling via selective autophagy. *Mol. Cell* **81**, 5082-5098.e11. doi:10.1016/j.molcel.2021.10.001

Ori-McKenney, K. M., Jan, L. Y. and Jan, Y.-N. (2012). Golgi outposts shape dendrite morphology by functioning as sites of acentrosomal microtubule nucleation in neurons. *Neuron* **76**, 921-930. doi:10.1016/j.neuron.2012.10.008

Pantazis, C. B., Yang, A., Lara, E., McDonough, J. A., Blauwendraat, C., Peng, L., Oguro, H., Kanaujiya, J., Zou, J., Sebesta, D. et al. (2022). A reference human induced pluripotent stem cell line for large-scale collaborative studies. *Cell Stem Cell* **29**, 1685-1702.e22. doi:10.1016/j.stem.2022.11.004

Park, J. H. and Roll-Mecak, A. (2018). The tubulin code in neuronal polarity. *Curr. Opin. Neurobiol.* **51**, 95-102. doi:10.1016/j.conb.2018.03.001

Polder, G., Hovens, H. L. E. and Zweers, A. J. (2010). Measuring shoot length of submerged aquatic plants using graph analysis. In *Proceedings of the ImageJ User and Developer Conference 2010, Mondorf-les-Bains, Luxembourg, 27-29 October 2010*, pp. 172-177. Centre de Recherche Public Henri Tudor.

Rhoads, S. N., Dong, W., Hsu, C.-H., Mfulama, N. R., Yarlagadda, R., Ragusa, J. V., Ye, M., Henrie, A., Zanellati, M. C., Diering, G. H. et al. (2025). Neurons and astrocytes have distinct organelle signatures and responses to stress. *Cell Rep.* **44**, 116280. doi:10.1016/j.celrep.2025.116280

Roll-Mecak, A. (2020). The tubulin code in microtubule dynamics and information encoding. *Dev. Cell* **54**, 7-20. doi:10.1016/j.devcel.2020.06.008

Schmoranzer, J. and Simon, S. M. (2003). Role of microtubules in fusion of post-golgi vesicles to the plasma membrane. *Mol. Biol. Cell* **14**, 1558-1569. doi:10.1091/mbc.e02-08-0500

Sébastien, M., Paquette, A. L., Prowse, E. N. P., Hendricks, A. G. and Brouhard, G. J. (2025). Doublecortin restricts neuronal branching by regulating tubulin polyglutamylation. *Nat. Commun.* **16**, 1749. doi:10.1038/s41467-025-56951-2

Steger, C. (1998). An unbiased detector of curvilinear structures. *IEEE Trans. Pattern Anal. Mach. Intell.* **20**, 113-125. doi:10.1109/34.659930

Valm, A. M., Cohen, S., Legant, W. R., Melunis, J., Hershberg, U., Wait, E., Cohen, A. R., Davidson, M. W., Betzig, E. and Lippincott-Schwartz, J. (2017). Applying systems-level spectral imaging and analysis to reveal the organelle interactome. *Nature* **546**, 162-167. doi:10.1038/nature22369

van Dijk, J., Rogowski, K., Miro, J., Lacroix, B., Eddé, B. and Janke, C. (2007). A targeted multienzyme mechanism for selective microtubule polyglutamylation. *Mol. Cell* **26**, 437-448. doi:10.1016/j.molcel.2007.04.012

Wang, J., Daniszewski, M., Hao, M. M., Hernández, D., Pébay, A., Gleeson, P. A. and Fourriere, L. (2023). Organelle mapping in dendrites of human iPSC-derived neurons reveals dynamic functional dendritic Golgi structures. *Cell Rep.* **42**, 112709. doi:10.1016/j.celrep.2023.112709

Zanellati, M. C., Hsu, C.-H. and Cohen, S. (2024). Imaging interorganelle contacts at a glance. *J. Cell Sci.* **137**, jcs262020. doi:10.1242/jcs.262020

Zanellati, M. C., Coman, Z., Bhowmik, D., Hsu, C.-H., Basundra, R., Rhoads, S. N., Mfulama, N. R., Ehrmann, B. M., Deshmukh, M. and Cohen, S. (2026). Organelle communication networks rewire to support lipid metabolism during neuronal differentiation. *bioRxiv*, 2026.02.10.704675. doi:10.64898/2026.02.10.704675

Zhang, S., Zhao, J., Quan, Z., Li, H. and Qing, H. (2022). Mitochondria and other organelles in neural development and their potential as therapeutic targets in neurodegenerative diseases. *Front. Neurosci.* **16**, 853911. doi:10.3389/fnins.2022.853911

Zheng, X., Boyer, L., Jin, M., Mertens, J., Kim, Y., Ma, L., Ma, L., Hamm, M., Gage, F. H. and Hunter, T. (2016). Metabolic reprogramming during neuronal differentiation from aerobic glycolysis to neuronal oxidative phosphorylation. *eLife* **5**, e13374. doi:10.7554/eLife.13374

Zheng, P., Obara, C. J., Szczesna, E., Nixon-Abell, J., Mahalingan, K. K., Roll-Mecak, A., Lippincott-Schwartz, J. and Blackstone, C. (2022). ER proteins decipher the tubulin code to regulate organelle distribution. *Nature* **601**, 132-138. doi:10.1038/s41586-021-04204-9

Zimmermann, J. A., Lucht, K., Stecher, M., Badhan, C., Glaser, K. M., Epple, M. W., Koch, L. R., Deboutte, W., Manke, T., Ebnet, K. et al. (2024). Functional multi-organelle units control inflammatory lipid metabolism of macrophages. *Nat. Cell Biol.* **26**, 1261-1273. doi:10.1038/s41556-024-01457-0